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Illegibly legible: Outcomes of a land records modernisation programme in South India

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Abstract

This article analyses an initiative in 2017 to update and digitise textual land records in Telangana, a south Indian state. Its premise is that experiments to modernise land records have not met with pre-determined standards of success due, primarily, to the historically evolved contradictions around land as a resource and a commodity. In countries like India, the colonial policies on land and the post-colonial success of landlords to manipulate land records had already left the post-War task of ensuring conclusive land titles intractable. In more recent times, however, the agenda of land records modernisation has been absorbed by neoliberalism, which aims to create free land markets and replace traditional subsidies with direct cash transfers. This article shows that, consequently, the task of land records modernisation in Telangana became disembedded from agendas of agrarian egalitarianism and was rendered more complex—historical errors and exclusions were reproduced in new ways; technocratic solutions of the bureaucracy made governmental processes opaque; landed sections continued to subvert implementation; tenants were excluded from the land titles; and there were fears that the scheme would be misused to ease corporate land acquisition. Land records modernisation remains important for agrarian reform, but its success remains contingent on a greater appreciation among policymakers for the historical and political economy aspects of land ownership and possession.

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KEYWORDS
land markets, land records modernisation, land reforms, land titles, *Rythu Bandhu*, Telangana

1 | INTRODUCTION

In societies where access to land is mediated by private property rights, updating and modernising land records is an important task of governance—with implications for rural equity and class and caste relations. Formalisation of land titles, accompanying or following land reforms, provides small and marginal farmers, tenants and the near-landless with secure tenure; it saves their time, reduces their costs of document storage and search and ensures document recovery in times of disasters. They improve the society's ability to reduce corruption in land administration systems—i.e., helps ensure 'transparency guarantee' (Sen, 1999). Even in societies that have not had land reform, formalisation of land titles helps in making the concentration of land ownership visible and help galvanise campaigns for land redistribution. Formal land titles can help tenants access bank credit (Karmakar et al., 2011). Above all, a modern land records system is important to maintain a robust statistical system and adopt scientific land use planning (Ramachandran, 2020).

Despite these obvious benefits, land records modernisation has remained a contentious policy project. On the one hand, contemporary policy processes around land records modernisation are increasingly aligned with the neoliberal goal of 'free land markets' and have shifted from the *developmental* arena to the *technocratic* sphere. On the other hand, efforts to modernise land records have been less successful than expected, invited resistance and subject to bureaucratic intransigence.

Our paper, then, begins with the following question: Do the sociocultural and political economy characteristics of land limit the possibilities of success in such schemes? The paper begins by laying down the arguments on the motivations for land records modernisation. It then documents and analyses one of the first large-scale efforts in India to update and digitise land records at the level of one state—Telangana. Telangana is a relatively new state in the Indian union, carved out of the erstwhile unified Andhra Pradesh State in 2014. The analysis is based on published literature, official reports, secondary data, primary data from the field, interviews with officials and farmers and data from a farmer-level helpline.

2 | THE CHALLENGE OF LAND ADMINISTRATION

2.1 | Changing conceptions of land

To begin with, land has always been a *site of production*. As human populations grew, the production of more food came to be associated with both extensive and intensive uses of land. However, the physical finiteness of land constrained its unmitigated extensive use, and the possession of land became a pre-requisite for controlling and regulating direct production (Marx, 1959). These constraints developed unevenly in different contexts and gave rise to varying forms of tensions around land as a resource.[1]

[1]As is well known, limitedness of land did not imply limitedness of land productivity; as Lenin (1964) pointed out, 'limitedness of the productivity of the land implies "limitedness" of the given technical level, the given state of the productive forces'. Technology, then, acted as a powerful intermediating force in the way tensions and conflicts around land evolved. If modern technologies relaxed the urgency to opt for extensiveness in production, they also gave a fillip to the possibilities of 'largeness' in land possession and use. The development of large-scale farming in the frontier regions became a possibility in the 19th century due to mechanisation, and the potential for trans-nationalisation of land possession and the attendant emergence of 'land grabs' expanded in the 20th century (see Byres, 1988; Levien, 2018).

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Later, debates developed around a second dimension: the *commodification* of land. The extensive and intensive uses of land as a resource coincided with the development of agrarian capitalism. Accumulation of capital centred around land as 'private property', which in turn was accompanied by increasing dispossession of people from land (Marx, 1959). After Marx, Karl Polanyi too noted how the commodification of land in market economies—disembedded from the society—gave rise to new contradictions. For Polanyi, land was eminently ill-suited to be a commodity; he termed the commodity description of land as 'entirely fictitious' (Polanyi, 1944).

The third dimension that emerged over time was a by-product of the commodification of land. As the demand for land increased, land was no more the site of agricultural production alone but also of multiple *non-agricultural uses*. Land was demanded for industries, services, housing, tourism, recreation and, above all, conservation. These competing uses tested the survival of land as a productive resource, intensified speculative activities in its purchase and sale and exacerbated conflicts around land (Sud, 2021).

Finally, none of the above tensions and conflicts within capitalist societies ameliorated the need for improving the *administration of land*. Modern forms of land administration, especially in maintaining land records and recording land transactions, became essential for the capitalist state to regulate the use of land as private property and contain the contradictions it gave rise to. As nation states developed, there also arose the need for uniformity in the design and architecture of land administration. Thus, over time, land administration came to be intrinsically linked to state formation, political processes, territorial disputes and economic policy.

The question we move ahead with, then, is the following: How did the need for 'efficient' land administration interface with the contradictions generated by the unique features of land?

2.2 | The challenge of land administration

Given the unique features of land, nation states have attempted to raise the efficiency of land use alongside the introduction of regulations to contain rising dispossession. In other words, governments have had the unenviable task of designing systems of land administration corresponding to both to the stage of development of productive forces and the changing nature of conflicts around land. The contemporary discourse around land administration is powerfully moulded by these twin tasks of modern-day governance.

In the post-War developing world, the task of land governance was additionally complicated by the refractory nature of land relations left over by colonialism. In colonies like India, the British empire broke down the traditional framework of land relations. It restructured land relations to maximise land revenues, created new landlords and vested them with legal powers of private property. But these new systems and records ended up generating fresh contradictions in the agrarian economy, as they were 'made not for the people, who cultivate the soil, nor for the holder, who owns it, but for the government that taxes it' (Marx, 1853).

Such contradictions, indeed, are the focus of Scott's (1998) argument that a new 'politics of measurement' emerged under the regimes of colonialism. The creation of land records led to the *simplification*, *codification* and *standardisation* of local customary practices around land. Scott refers to this process as a shift from the 'illegibility' to the 'legibility' of traditional communal land tenures. In particular, the preparation of cadastral maps allowed the colonial states to abstract and universalise. These efforts sought to transform and flatten the diversity in customary land regimes; concurrently, they also gave rise to movements of anti-colonial resistance.

Even after colonialism ended, land administration in most erstwhile colonies was locked into trajectories of 'path dependence'. Other than a few societies that transformed agrarian relations through land reforms, colonial systems of land administration survived (see Byres, 1988). To escape provisions of land ceiling laws and evict tenants, landlords and the revenue bureaucracy colluded and extensively manipulated land records.

Consequently, land records lost their correspondence with the reality on the ground. This lack of correspondence has led to much flailing of the existing systems of land administration. 'Illegible' land records—where a series of traditional and informal set of tenures and settlements survived over many layers of claims and counterclaims, and

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piles of editing and overwriting—led to disorders in defining property rights. Thus, the task of land administration came to be identified as one of transforming these 'illegible' records into 'legible' records. Essentially, it was this outstanding task of the post-War state that was redefined by neoliberalism in the 1980s.

2.3 | Land records in the neoliberal era

From the 1980s and after, the task of 'modernisation of land records' became more prominent in policy discourse. Neoliberal economists, international organisations and policy consultants argued that land records must be modernised not to transform agrarian relations but to create a free land market, where land would be freely transferable and new investments in land would be facilitated (see Byamugisha, 1999; de Soto, 2000; Feder & Nishio, 1998; McKinsey, 2001). They argued that *conclusive* land titles improve the Ease of Doing Business (EoDB).[2] The World Bank's EoDB rankings were used to argue the case for conclusive land titling (NCAER, 2020).[3] New solutions like block chain were also prescribed to create a robust property rights system that facilitated investment in land (see Bal, 2017).

Thus, in the 1980s, bilateral and multilateral institutions, such as the United States Agency for International Development (USAID), began to fund land titling projects in developing countries (Deininger & Feder, 2013). In the 1990s, the World Bank assisted governments in 11 former Soviet Bloc countries to develop and implement land registration programmes. By 2005, the World Bank was supervising a portfolio of land administration projects worth more than $1 billion (Bell, 2006; see also Nayak, 2020).[4]

Neoliberalism in land administration was closely associated with the rise of 'technocracy' in policy. Technocracy is defined in many ways—as the replacement of 'government of persons' by 'administration of things'; as a 'government by experts' necessitated by the incapability of citizens to self-govern; as 'the rise to power of those who possess technical knowledge or ability, to the detriment of the traditional type of politician'; and as 'the administrative and political domination of society by a state elite … through the application of instrumentally rational techniques' (Cole, 2022). But essentially, technocracy implies a depoliticisation of policy, actively intermediated by state bureaucracies, which tends to insulate the state from democracy to promote a set of efficiency goals. The rise of technocracy does not just maintain, if not deepen, elite domination; it fetishises the need to contain political pressures and legitimises the autonomy of 'experts'.

There is little excitement in the literature on the robustness of EoDB indices, even within institutions like the World Bank (Nagaraj, 2019). According to Nagaraj, 'there is no statistically valid association between the EoDB index rank … and growth in FDI and private investment'. Given the many controversies surrounding the estimation of EoDB ranks, the World Bank itself discontinued its publication after 2020.[5]

The empirical literature on land titling and its relationship with land values, equity and investment is also ambiguous on whether conclusive land titling leads to economically 'efficient' outcomes. Miceli and Kieyah (2003) reviewed the evidence for the United States where many states attempted a shift from a recording system to a voluntary Torrens-based land registration system. They argued that the policy was only successful in 'solving idiosyncratic problems in the record' and did not lead to a reduction in costs of transacting land. Even in the most successful states, only less than half of the records entered the new system.

[2]The concept of *conclusive* titling over land is also called the *Torrens system*; it was named after Sir Robert Richard Torrens, who established it in southern Australia in 1858. The Torrens system insists on three conditions: (a) the *mirror* principle (i.e., the document would mirror the truth on the ground); (b) the *curtain* principle (i.e., the document represents a closure of all disputes prior to its date of issue); and (c) the *indemnity* principle (i.e., the document is equivalent to title insurance, qualifying its holder to a compensation in case of any error).

[3]Other scholars, even while agreeing to settle in for *presumptive* land titling, advocate the need for a system of private land title insurance that would, even in the absence of conclusive titling, make land more 'marketable' (see Burman, 2019; Gupta, 2011).

[4]The logic, here, was also based on the theories of new institutional economics (NIE). Neoclassical economic theories had assumed the presence of perfect and costless information, leading to zero transaction costs. However, new institutional theories were built on the assumption that there were asymmetric information and transaction costs, which must be minimised for efficient choices and free flow of capital (Byamugisha, 1999). It was held that a system of conclusive land titles minimised transaction costs.

[5]See 'World Bank Group to Discontinue Doing Business Report', World Bank, at https://bit.ly/3EtOkhi.

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An early review of the African efforts to transform customary land systems in sub-Saharan Africa argued that land titling or registration 'would not have the intended impact, would not be economically justifiable, or would even be counter-productive' (Atwood, 1990, p. 668). They fail to account for local and broader national class relations (as, for instance, described in Capps, 2016). In his work, Atwood also remarks that the theoretical framework on free land markets failed to 'take adequate account of the extra-legal, informal, local institutional environment' within which most rural households accessed land. The case of the Brazilian frontier discussed in Alston et al. (1999) provided similar proof of why title reforms may fail. Further, even if distortions are removed in the land market, distortions that survived in other markets, and market failures therein, may defeat the purpose of land title reforms (Dong, 1996).

An alternative perspective is that land records modernisation in rural areas, viewed from within the EoDB framework, seeks to facilitate large-scale corporate investments in land and agri-business (see Mittal, 2014). The example of a state-level initiative launched in India in the state of Karnataka—titled *Bhoomi*—is a good example.

Launched in 2000, the *Bhoomi* initiative aimed to computerise land records and provide online delivery of revenue documents. It was not based on a fresh resurvey of land. The Record of Rights, Tenancy and Crops (RTC) was digitised, that is, entered in a portal on an 'as is what is' basis, and then subjected to preliminary verification. *Bhoomi* was also a manifestation of technocracy, the argument being that illegible land records reduced efficiency and fostered corruption. However, as Benjamin and Raman (2011) argued, the premise of *Bhoomi* was to allow big businesses to locate in the rapidly urbanising peripheries of Karnataka's big cities like Bengaluru. The scheme became a tool to reshape power relations around land on new terms while erasing much of the traditional claims over land. The problem, according to them, was in seeing the illegibility of land records not as a 'political condition' but as a 'normative' problem (see also De, 2016).[6] Such evidence that questions the faith of neoliberalism in creating free land markets has not slowed down governmental efforts in that direction.

More recently, there is an additional reason why the presence of accurate land records has become significant. The increasing reliance on cash transfers, which crowd out funding for traditional in-kind schemes in different social and economic sectors, has dovetailed the growth of neoliberalism in countries like India. Eligibility to access the benefits of specific government programmes is linked to the possession of accurate land titles. For example, under the programme of liberalisation in agriculture, fertiliser subsidies are being monetised and replaced with direct cash transfers. The quantum of these cash transfers is linked to the extent of land owned; thus, a farmer must possess a clear land record to be eligible for the transfer (Ramesh, 2020; Uday & Zaveri, 2020).

In May 2018, the Telangana government introduced a scheme called the *Rythu Bandhu Scheme* (RBS). The RBS involved a direct cash payment to each farmer—an 'agricultural investment support' of Rs 5000 (USD 60) per acre per season—where only those who possessed accurate land titles were recognised as farmers. [Correction added on 18 September 2023, after first online publication: dollar conversion for 'Rs 5000' has been provided in the preceding sentence.] As land records were outdated, the Telangana government initiated a scheme to update textual land records in September 2017. The programme was called the Land Records Updation Programme (LRUP). Its objective was limited: fix private property rights and use it to identify the beneficiaries of the RBS. Nevertheless, the outcomes of LRUP remained vulnerable to the systemic problems that beset other land records modernisation programmes driven by a neoliberal persuasion.

This brings us to the more specific questions that we try to probe in this paper, drawing from a case study of the LRUP. Can the prospects of land records modernisation be divorced from the political economy of land ownership and possession? Can technocratic solutions bypass the complexities in land systems inherited from colonialism and ensure 'legibility' in land titles? Can land records modernisation be conceived of as autonomous from the power relations that are historically embedded in land ownership and possession? What do programmes that seek to dis-embed claims over land from history, and re-embed them in modern notions of ownership and possession, imply for specific agrarian groups like tenants, labourers and tribal households?

[6]As De (2016, p. 24) points out, issues of land titling in India are historically intertwined with 'local knowledge and practices, the fuzziness of customary land rights, the dynamics of interests within local structures of power and the compulsions of electoral calculations intrinsic to a dynamic federal polity'.

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3 | LAND RECORDS IN INDIA: COMPLEXITY AND THE CHALLENGE

3.1 | An introduction to land records

Land is a state subject under the Indian constitution, and land records are maintained by the state governments. Land records are official documents that contain information about each parcel of land. The information contained in the documents include the extent of area, its boundaries, the owner of the property, nature of rights of its occupants, nature of use and details of sale and purchase. There are several types of land records that contain distinct sets of information about a land parcel. As such, land records can be classified into three: (a) *textual records*, maintained by the Revenue Departments; (b) *spatial records*, maintained by the Survey and Settlement Departments; and (c) *transaction records*, maintained by the Registration Departments (see Table 1 for a summary).

These three departments have historically worked in compartments or silos. Data collected or recorded by one department are not necessarily reflected in the records for the same plot maintained by other departments. We shall provide some typical examples. First, when the sale of one plot of land takes place by executing a deed in the Registration Department, the information on change in ownership is not reflected as a change in the Record of Rights (RoR) maintained by the Revenue Department. Instead, the list of deeds executed is maintained in a separate

TABLE 1 A summary of the contents of various types of land records maintained in Indian villages.

Department and function	Types of documents maintained (names differ across states)	Short description
The Revenue Department	Record of Rights (RoR); Form 1-B in Telangana	The RoR is the primary land record and is equivalent to a title to the land. It records the nature, limitation and extent of the respective rights and interests of all parties and information on encumbrances, such as mortgages on the land, rent and revenue.
	Mutation register	A mutation register is a list of changes in the entries in the RoR due to a sale or purchase or death. This information is sent by the Department of Registration.
	Adangal/Khasra Girdawari/Khatian/Khasra; *Pahani* in Telangana	The *pahani* is a document with the survey and parcel number for each plot of land with the owner's name, details of area, cropping pattern, soil type and sources of irrigation. It is maintained on a seasonal basis to compute land taxes.
	Khata register	The *khata* is a land tax record that shows land revenue, cess, water rates, other government dues to be paid and the amount paid each year.
The Survey and Settlement Department	Field Measurement Book (FMB); *Tippan* in Telangana	The FMB records every plot's relative coordinates, boundaries and distance measurements in chains, links or another local unit. Each plot has a unique survey number.
	Village/cadastral maps	The information recorded in the FMB is used to compile the village/cadastral map. This is usually undertaken based on a smaller measurement scale.
The Department of Stamps and Registration	Sale deed	The sale deed is a contract signed and registered on a judicial stamp paper by the seller and buyer of a land plot. It contains the names of the buyer and seller, area of land and price paid.

Note: The table provides only a broad overview of the documentation undertaken and maintained. The name of each document varies across states, and the table does not provide all the names used in every state for each type of record.
Source: Collated from multiple sources, including official records, literature and interviews with revenue officials.

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mutation register until a resurvey leads to the finalisation of a new RoR. However, the last resurvey in most Indian villages was conducted during the colonial period. The average age of village/cadastral maps in most states was estimated in 2009 at more than 50 years (GoI, 2009).

Second, when a plot with one survey number (say, 259) is divided into two plots due to sale or sub-division, two new survey numbers are generated (say, 259-A and 259-B) for each plot. However, the RoR would continue to carry the old undivided survey number (i.e., 259). It is updated with new survey numbers only after a resurvey is completed.

Third, in many cases, the name of the 'owner' of a plot of land, as listed in the RoR, is the name of a person who may have been deceased for many years, if not decades. Due to differences within the family on sub-division or due to the reluctance to pay stamp duties, ownership details are left unchanged in the official registers; that is, no new deed is executed to effect a change in ownership. On some occasions, even intervening transactions, such as heirship partitions that do not require to be registered, are not recorded. On other occasions, a large plot of land is jointly owned by more than two brothers, but each of them cultivates distinct parcels as if they own them.

In sum, the system of land records in most Indian states is outdated, ad hoc and fragmented across several departments and registers (see also Alm et al., 2004; GoI, 2009; Narayanan et al., 2017; Wadhwa, 2002; World Bank, 2007). It is in this context that the objective of land records modernisation was seen as relevant for policy.

3.2 | A review of past efforts

In India, the need for updated and accurate land titles was recognised as important in most Five-Year Plan documents from the 1950s (Wadhwa, 2002). The interest in land titling in the 1950s arose from the need to implement land reforms. As is well documented, land reforms in India were a failure. According to Appu (1996), the neglect of land records in India after independence was closely associated with the efforts of landlords to evade land reform laws; in this sense, the poor quality of land records had the implicit sanction of the Indian state. For example, tenants were systematically omitted from mention in the land records, which allowed landlords to evict them. Land records were also regularly manipulated to ensure that landlords escaped the provisions of land ceiling laws (see also Srivastava, 2015).

In 1973, a Task Force of the Planning Commission observed that 'the absence of up-to-date land records is a serious obstacle in the implementation of land reforms' (Planning Commission, 1973). The Panel on Land Reforms in 1961 and the National Commission on Agriculture in 1976 made similar observations (GoI, 1961; NCA, 1976).

With the end of official interest in land reforms by the 1980s and the initiation of economic liberalisation policies in the 1990s, the motivation for modernising land records changed qualitatively. In the new era, modernisation of land records came to be linked to assured private property rights and the improvement of business confidence (Nayak, 2020). In 1987, the Planning Commission headed by Manmohan Singh, who inaugurated India's liberalisation programme as Finance Minister in 1991, appointed a one-man committee headed by D. C. Wadhwa to study the problem (Wadhwa, 1989). This committee recommended that India should shift from a *presumptive* land titling system to a modern and computerised *conclusive* land titling system.[7] After the Wadhwa committee report, India conducted the first pilots on computerising and updating land records in Bihar and Odisha in 1988. In 1989–1990, a similar initiative was launched in all states (Panicker, 2015). However, the implementation was poor.

In 2008, two surviving schemes in this realm—viz., 'Strengthening of Revenue Administration and Updating of Land Records' and 'Computerisation of Land Records'—were merged into a new 'National Land Records

[7]Given the complexity of land systems and the multiplicity of poorly inter-linked records, India has effectively followed a system of *presumptive titling*. A presumptive title implies that a registered deed can at best provide *evidence* of one's claim to a property and does not offer *conclusive proof* of ownership or a government guaranteed title to property (Goswami et al., 2020). Given that registered deeds prevail over land titles and registrars do not check revenue records before registration, deeds are frequently contested in courts of law (Wadhwa, 2002). In a survey in 2015–2016, it was estimated that 66.2% of all the civil litigations in the district courts of India pertained to land- or property-related cases (Daksh, 2016).

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Modernisation Programme' (NLRMP). As per the NLRMP, land records in all districts were to be computerised and updated by 2017 using a Public-Private-Partnership model (Bal, 2017). This scheme also did not make much headway and, consequently, a revamped NLRMP was introduced in 2014 as the Digital India Land Records Modernization Programme (DILRMP). The DILRMP had four components: (a) digitising and updating existing textual records; (b) converting existing maps into digitised maps; (c) resurvey, resettlement and registration; and (d) merger of land ownership data with land use data. The DILRMP began as a centrally sponsored scheme with shared central and state government funding. However, it was converted into a central sector scheme in 2016 with 100% funding from the centre (Narayanan et al., 2018).

The progress of DILRMP, as of September 2022, is summarised in Table 2. Data in Table 2 give the impression that progress under DILRMP was considerable. However, much of the computerisation completed was of data from the RoRs prepared during the colonial period. The mutation registers were computerised separately but not integrated to the RoR everywhere, either *post-facto* or in real time. A fully integrated RoR required a resurvey; but till 2022, only about 13% of the villages in the country were resurveyed, and only about 9% of country's area was resurveyed.

An impact assessment of DILRMP in the states of Himachal Pradesh, Rajasthan and Maharashtra showed variations between on-the-ground situation and what was contained in the land records (Narayanan et al., 2017; NCAER, 2017). For instance, in the two study sites in Himachal Pradesh, the average share of land area with more than 5% variation across the RoR and on-the-ground situation was 65%. The corresponding share was 79% in Maharashtra and 68% in Rajasthan. In Maharashtra, about 50% of the sample land parcels recorded a more than 20% deviation in area from the revenue records. Only <4% of the total cadastral maps were digitised. Further, the inter-connectivity between the Registration Department and the Revenue Department was marred by server breakdowns and power shortages. These depressing results were broadly illustrative of the national situation on land records modernisation.

3.3 | LRUP in Telangana

The LRUP in Telangana followed the broad guidelines of the DILRMP, but it retained its specific motivation: create initial conditions for the RBS. However, the complexities in the history of land relations in Telangana had left behind

TABLE 2 Progress in selected components of DILRMP, India, as of September 2022, in percentage.

Components of land records modernisation	Progress (%)
Percentage of villages that have completed more than 95% of computerisation of land records	94.0
Percentage of villages that have completed more than 95% of computerisation of mutation registers	80.9
Percentage of villages that can issue digitally signed RoR	64.0
Percentage of villages that have completed more than 95% of spatial data verification	68.5
Percentage of villages that have completed more than 95% of work on linking cadastral maps and RoR	60.7
Percentage of villages that have completed more than 95% of work on real-time updating of RoR and maps	29.8
Percentage of villages that have completed resurvey	13.4
Percentage of cadastral maps that have been digitised	70.4
Percentage of total area surveyed or resurveyed in India	8.6

Abbreviations: DILRMP, Digital India Land Records Modernization Programme; RoR, Record of Rights.
Source: Department of Land Resources (DoLR), Government of India.

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a legacy of unreliable land titles. According to one estimate in 2017, there were 4.2 million 'errors' in Telangana's land records.[8] Further, highlighting the relationship between land ownership and tenancy, the Agriculture Department and the Revenue Department had different figures on the number of 'farmers'.[9]

LRUP was positioned by the government as essential to remove these errors and create a 'pure' database of RBS beneficiaries. The exercise was to run from September 2017 till December 2017 although the scheme ran till April 2018 in many districts. The objectives of the LRUP—described by the state government as 'a special time-bound, purification and updation of land records'—were twofold.[10] First, it aimed to generate accurate land records in Telangana, which would reflect the ground reality. Second, it aimed to establish a transparent and accountable land records management system (LRMS).

To analyse the progress of the LRUP, we adopted two sets of methodologies in 2019. Although we tried to understand how the scheme operated at the state level, we also tried to focus on one district—Adilabad—to study the operation of the scheme at the lower levels of the revenue administration. First, we interviewed revenue officials at each level of the organisational hierarchy to understand and document their contribution to the process. At least two officials at each rung of the administrative hierarchy were interviewed. Besides interviewing officials at the office of the Chief Commissioner of Land Administration (CCLA) at the state level and the collectors at the district level, we conducted semi-structured interviews with revenue divisional officers (RDOs) at the division level; mandal revenue officials (MROs) or *tehsildars* at the *mandal* level; and village revenue officials (VRO) or *patwaris* and revenue inspectors (RIs) at the village level. Questions were asked about the objectives of LRUP, how the exercise was undertaken, how responsibilities were fixed, the pattern of reporting systems, the timelines and how disputes were resolved.

Second, we specifically focused on studying the implementation of the scheme in the Talamadagu *mandal* in Adilabad district, which is the northernmost district of Telangana. Our access to the field was facilitated by the *Rythu Swarajya Vedika* (RSV), a Hyderabad-based organisation working on farmer's rights. We spoke to a range of revenue officials in Adilabad at multiple levels and often snowball-interviewed other officials that were suggested to us for specific clarifications. We reviewed government records on the LRUP at the district level and below. Finally, we used anonymised data from a farmer's complaint database, which was put together by the RSV through a widely publicised helpline number. This database allowed us to record and classify the types of complaints regarding LRUP.

4 | TELANGANA'S LAND RECORDS: THE COLONIAL LEGACY

The unreliability of land records in Telangana in 2017 was an outcome of more than a century of changes in the land revenue system. The accumulation of changes in the land revenue system, and the nature of landlord-led manipulation of land records after the 1940s, must be documented to better appreciate the links between agrarian change, class relations and land titling in the region.

During the colonial period, the present-day states in India belonged to different regimes of British land administration, such as *zamindari*, *ryotwari*, *mahalwari*, *khatedari*, *taluqdari* and *malguzari*. In addition, there were princely states that followed distinct systems of land administration. Given the differences in colonial land systems, the contemporary systems of land administration also vary across states.[11] Even within one state, there exist multiple

[8]'Telangana: Nizam-era Land Data Has 42 Lakh Errors', *Deccan Chronicle*, 19 August 2017, available at https://bit.ly/2BU68a9, accessed on 26 July 2020.
[9]This was evident from official reports of the Chief Commissioner of Land Administration (CCLA) we reviewed.
[10]See Circular No. 1, Ref. No. CMRO/342/2017, 'Land Records Updation Project: Purification and Updation of Land Records—Detailed Guidelines', issued on 09-09-2017, Office of CCLA, Government of Telangana.
[11]The primacy of revenue collection in the colonial period led to a system of land records that focused on *use* rather than *ownership*, creating limited incentives for owners to update land records. This was especially so in the *zamindari* regions, where land documents provided only those details pertaining to the need of land revenue assessment. In the *ryotwari* regions, because revenue collection directly involved the cultivators, the RoRs were more carefully maintained and were more detailed than in the *zamindari* regions (World Bank, 2007).

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methods and terminologies in the maintenance of land records. A good example is the difference between the Andhra and Telangana regions in the erstwhile unified state of Andhra Pradesh.

Telangana has had a complex legacy of land records because the region inherited multiple systems of land administration from the colonial period. Before the 18th century, Hyderabad State was part of the Mughal Empire. In 1724, Asaf Jah Nizam-ul-Mulk I declared independence from the Mughals and established the Hyderabad State. For a while, the Nizam continued with the land records and settlement systems used by the Mughals. Over the years, however, revenues from land declined. By the early 19th century, Hyderabad was close to bankruptcy due to more than one reason: the Treaty of Subsidiary Alliance between the Nizam and the British and Nizam's heavy debts to the Palmer and Company, the Rohillas and the Arab merchants (Bobbili & Manohar, 1999, 2001).

In the later part of the 19th century, the then Nizam introduced changes in the land revenue system to bolster land revenues. In 1866, he abolished the earlier revenue system, eliminated intermediary revenue collectors and established direct contact between the state and the peasant (ibid.). In 1875, a new land settlement was put in place. In 1907, the Land Revenue Act 1317F was passed. Between 1932 and 1934, a fresh settlement survey was conducted. This 'survey settlement and *bandobust* operation' of 1932–1934 is considered by today's revenue authorities as the 'most authentic' source of land records in Telangana. It produced the *sethwar*, a new settlement register.

Although these legislations and measures were primarily aimed at raising land revenues, it also facilitated the concentration of land in the hands of landlords. During this period, of all land in Hyderabad State, 60% was under the governmental land revenue system, called *diwani* or *khalsa* area. About 30% was under the *jagirdari* system, and about 10% was Nizam's direct estate, or *sarf khas* (Sundarayya, 1972). It was in the *jagirdari* and *sarf khas* areas that the exploitation of tenants and labourers was most intense. A major portion of the land previously cultivated by tenants was captured by the landlords, who got lands registered in their names, sometimes even without the knowledge of the tenants.

The heavy exploitation of the peasantry resulted in the famous Telangana struggle led by the Communist Party of India in 1946. In 1948, Hyderabad was annexed to the Indian Union. In 1950, the Hyderabad Tenancy and Agricultural Lands Act was passed as a response to the Telangana struggle. This Act abolished the *sarf khas* and *jagirdari* systems and merged them with the *diwani* areas. It also introduced a document called the 38-E certificate, which accorded protection to tenants holding land for a period of 6 years and gave them permanent rights to remain as tenants subject to payment of rents and not sub-letting to others (Khusro, 1958).

With the 1950 Act and the abolition of *sarf khas* and *jagirdari*, a new responsibility of the government was the identification of the actual cultivators and recording their names in the revenue records. For this purpose, the preparation of the *khasra pahani* began in 1954 (Bobbili & Manohar, 1999, 2001). In the *khasra pahani* document, *khata* numbers were to be given to landholders, with the extent of land in possession and survey numbers allotted to the plots. The *khasra pahani* was to have four components: the *bandobust* record (details of survey settlement), RoR, agricultural accounts and cultivation details. Curiously, no new survey was undertaken prior to the preparation of the *khasra pahani*. Instead, the old *sethwar* was used and a mutation register was prepared with the history of possession and transaction of each plot after 1934 (such as purchase, inheritance, partnership, maintenance or gifts). The name of the existing holder was recorded as the *pattadar*. In many cases, the *pattadar* was a landlord or a large landowner, and hence, the names of his/her tenants were also to be recorded. However, no details of the location or extent of the lands of tenants were entered. This, later, became a major anomaly in the land records.

After the *khasra pahani* was prepared in 1954, the Revenue Department demanded that each village send information on actual land possessed and cultivated. Under pressure from landlords, the village *patwaris* sent the department the list of original holders listed in the *sethwar* and not the names of actual cultivators. Based on this faulty list sent by the *patwaris*, the Revenue Department prepared a new *vasoolbaki* register, which consisted of the old survey number with the area of land and the new survey number with the area of land. Given the way the *vasoolbaki* register was prepared, the total area of land in the *sethwar*, *khasra pahani* and *vasoolbaki* registers of the same village did not match in most cases. Over time, this emerged as another anomaly in the land records.

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In 1971, the Andhra Pradesh Record of Rights in Land and Pattadar Passbooks Act was passed. This 1971 Act prescribed a format for the RoR and mandated the issue of a new title record to each landholder called the *Patta* Passbook (PPB). In 1973, the Andhra Pradesh Land Reforms (Ceiling on Agricultural Holdings) Act was passed. When the *sethwar* (updated from 1932–1934 to 1955) was shared with the villages to implement the 1973 Act, another series of manipulations in land records began, particularly to exclude the names of tenant farmers. The name of the original *pattadar* was to be changed with the name of the actual tiller, but instead, the names of *benamis* of landlords and large landowners were inserted. In addition, new survey numbers were added against their names. The old survey numbers could no longer be realistically compared with the new survey numbers, as the areas of land changed with the names of the possessors. In many cases, landholders were not even aware of these changes in land records. As a result, the *khasra pahani* prepared in 1954 had little comparability with the *khasra pahani* prepared after 1973.

In sum, no survey of land took place in Telangana after 1932–1934. Most registers prepared after 1932–1934, especially in the 1970s, were products of manipulation engineered by landlords—to evade land ceiling laws—with the connivance of revenue officials and the *patwaris*.

In summary, the system of land records inherited by Telangana in 2017, which emerged from the Nizam's rule before 1948 and later the united state of Andhra Pradesh, presented a state of disorder. An official report in 2016 listed the ways in which the reality on the ground was different from the information contained in official land records (CAG, 2016). A pilot governmental exercise showed the major reasons for the divergence: irregular mutations in land records; unrecorded partitions; errors in recording names, extent and survey numbers; duplication of *khata* numbers; issues with *inam* land; and details of government lands, forest lands, *wakf* land, *bhoodan* land and endowment land.[12] Here was a typical case of 'illegibility' inherited from the past, where a maze of historical socio-legal relationships manifested in a 'heterogeneity of tenure forms' and a 'bundle of competing claims' in land records (Benjamin & Raman, 2011).

5 | THE IMPLEMENTATION OF LRUP

The implementation of LRUP was organised into two parts: The first part (Part A) involved identifying land parcels with either no disputes or undisputed errors (such as errors in names, surnames, extent of land and *khata* number, and the deletion of portions of land already sold). The second part (Part B) involved setting aside records with unresolved disputes or pending court cases.

5.1 | The administrative process

The first aim of LRUP was to update the PPBs of landowners and converge the information in the PPBs and the RoR. The second aim was to update the information on all government land, forest land, *wakf* land and endowment land. The third aim was to issue new PPBs with security features like QR codes, bar codes and the *Aadhaar* number. Detailed training sessions were organised for district collectors, *tehsildars*, deputy *tehsildars*, senior assistants, junior assistants, RIs and VROs. Publicity was ensured through vehicles fitted with mikes, local television channels and newspapers. At the state, district, *mandal* and village levels, a *Rythu Samanvaya Samithi* (Farmer's Coordination Committee) was set up to identify issues and intimate the concerned officers. Table 3 provides some basic details of LRUP's coverage.

Each team was headed by a *tehsildar*, a deputy *tehsildar* and an RI. The team was to collect and carry a set of documents before going to each village: (a) *sethwar/khasra pahani* or the oldest available *pahani* as the base record; (b) the latest *pahani*; (c) the 1-B register or the RoR; (d) the amendment registers for the preceding 10 years; and (e) a

[12]Interview with officials of the Department of Revenue, Government of Telangana, 2019.

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TABLE 3 Details of the coverage of LRUP, Telangana, September to December 2017 (extended to March 2018).

Duration	Coverage
Number of districts covered	30
Number of *mandals* covered	568
Number of villages covered	10,283
Number of teams deployed	1523
Number of revenue officers deployed	3500
Approximate total area verified (acres)	23,800,000
Area covered by one team per day (acres)	250
Number of villages assigned to each team	9
Number of days per village	10
Number of days per district	100

Abbreviation: LRUP, Land Records Updation Programme.
Source: Chief Commissioner of Land Administration Office, Revenue Department, Hyderabad.

village map. Additionally, each team also had the list of probable corrections in the names of *pattadars* and extent of area, particularly in the reconciliation of areas across the old and new *pahani* registers. For a survey number, the area noted in the original *sethwar/khasra pahani* was the benchmark against which corrections of area in the most recent *pahani* were to be made. The teams were also to carry a list of the civil and revenue cases for each parcel of land in their internal registers.

A timetable of the team visits was distributed in advance to all *pattadars* in the villages. Work began with the *patwaris* distributing 1-B extracts in the RoR to every household. Each household was given instructions to assess the 1-B extract and corroborate it with the details in their PPBs. They were given 8–10 days to study the two documents. After 8–10 days, the first *gram sabha* was held where villagers were asked to report discrepancies between the two documents. The team would then proceed to the next village. After all villages in a *mandal* were completed, a final *gram sabha* was convened. There, the final 1-B list was released, and people were informed about how the disputes were resolved.

The revenue teams were also supposed to undertake survey number-wise reconciliation of the actual area with the corresponding figures in the *sethwar/pahani*, maintain an acknowledgment register for the distribution of 1-B extracts and file the minutes of the first and last *gram sabhas*. They also had to keep a file on the claims and objections received during the field visits. At the end, each team was to declare that the RoR of that village was updated and made error-free to the maximum possible extent. A final copy of the RoR with the signatures of all landowners was to be published in the village *chavdi*.

At the end of the whole process, each district collector was required to compile final village-wise reports. The reports included (a) variations in area between *sethwar* and *pahani*; (b) details of unsettled or disputed issues in the villages; (c) details of missing survey numbers; and (d) details on non-agricultural land and government assigned land.

5.2 | Disputes and the process of resolution

In this sub-section, we discuss the kinds of disputes we identified in the field and the different methods of resolution. The variety of disputes we identified were evidence to the accumulation of historical factors that shaped ownership, possession and use of land in Telangana.

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a. Disputes arising out of deaths (*fouthis*), successions and partitions

In many cases, the land was possessed and used by the grandchildren, but the 1-B register in RoR continued to carry the grandparent's name. In other words, registrations of death and the subsequent partitions pertaining to the land parcel were never carried out. This was because the Registration Act of 1908 mandated the registration of partition during the lifetime of a landowner but heirship partitions after death were not required to be registered. In such cases, the revenue teams drew a chain of ownership and transfer to trace the original owner and from there back to the existing owner. As the collector of Adilabad explained to us in an interview:

> *There are multiple sub-divisions in most of the initial survey numbers. Let us say there is a person with 10 acres of land with survey number 250. He divides the land equally amongst his five sons and the consequent sub-divisions of the survey number are 250/1, 250/2, 250/3, 250/4 and 250/5. Today, 250/5 is possessed by the fifth son T. But the 1-B register shows that the plot is owned by the dead father. So, verification begins by drawing a family tree. If T claimed five acres of land (instead of two acres), we enquired about the other four brothers and to what they were staking claim to. Local enquiries, including an enjoyment survey, were also carried out if necessary.*

b. Disputes arising out of mutations

Let us consider the sale of a plot by A to B. B currently held the PPB, but the 1-B register of RoR continued to show A's name. These cases arose due to poor communication between the Registration Department and the Revenue Department. These were resolved by a concurrent reference to the documents of both departments and updating the respective record.

c. Disputes arising due to *sadabainama* transactions

Sadabainama transactions are unregistered transactions on white paper. Transfer of land titles through such transactions, despite not having legal sanctity, was popular in Telangana. Many people who had purchased land through *sadabainama* transactions did not have *pattas*. Under LRUP, small and marginal farmers (i.e., with land below 5 acres), who purchased the land in rural areas on or before 2 June 2014 and applied before 31 December 2017, were eligible to obtain a 13-B certificate (i.e., a document equivalent to a *patta*). A landowner with a 13-B certificate was considered a legal landowner. In such cases, the revenue staff conducted an enjoyment survey, checked from neighbouring landowners, perused documents that were presented to them, received concurrence from the *gram sabha* and issued the certificate.

d. Disputes arising due to encroachments in government assigned land

During the LRUP, the government decided to ensure secured land rights over encroached land to the 'eligible poor' or *sivai jamadars* (eligible and regularised). This was possible if the possessor of the encroached land belonged to the 'Below Poverty Line' category and had <5 acres of land. In such cases, fresh 1-B extracts were distributed to the occupants.

e. Disputes arising in acquired lands

Land is regularly acquired by the government for developmental or other reasons. However, in some cases, the PPB with the erstwhile landowner continued to mention the old owner's name. Such records were appropriately updated during the field visits.

f. Disputes in *wakf* or endowment or *bhoodan* land

The revenue teams were asked to identify and specifically survey *wakf* or endowment or *bhoodan* lands to ascertain if any other party had possession. There were three separate legislations governing these lands. The revenue team did not pass a judgement on these cases. These cases were directly classified as Part B.

g. Disputes related to name, area and survey number

In many cases, the spelling of names was entered wrongly across records. In other cases, the name of landowner did not match across the *Aadhaar* database or PPB or other documents. Disputes related to the area under a survey number emerged when the RoR and *pahani* had different figures for area under the same survey number. In

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other cases, two or more land parcels were consolidated into a single parcel in the past, but discrepancies survived across the registration records, revenue records and the entries in PPBs. The *tehsildar* of Tandur *mandal* gave us an example to describe how such disputes were settled:

> *Suppose the sethwar/khasra pahani/the oldest available pahani mentions the extent of a survey number, owned by A, to be 10 acres and 20 guntas (40 guntas = one acre). But the pahani of 2017 shows it as 11 acres. We try to find out how 20 guntas got added. One possibility is that a sale deed for the 20 guntas was executed between A and B in 2005. However, due to poor coordination between the revenue and registration departments, A continues to be the owner of the 20 guntas in the revenue records. That is, there is double entry of those 20 guntas. A can also claim that the sale never took place, or the sale was called off. So, we check the pahanis and 1-B for every year. We find that A owned 20 guntas in 2005, but A is missing in 2006 from the pahani. We check the sale deeds in 2005 and 2006 to assess whether a sale took place. We also check the cultivator column in the pahani and enquire from the neighbours whether A continued to cultivate the land.*

h. Disputes in Schedule 5 areas and forest lands
The Andhra Pradesh Scheduled Areas Land Transfer Regulation Act of 1970 (the '1/70 Act') restricted the transfer of tribal land from tribals to non-tribals (Rao, 2014). Cases were found where a non-tribal person was found in possession of land in a Schedule 5 area from before 1970 but did not possess documents to establish the claim. There were also forest enclosures (i.e., non-revenue land) that were recognised as revenue land under section 4 (1)(h) of the Forest Rights Act (FRA), 2006, of the Government of India. Such areas were marked as disputed and classified under Part B. A team of officials from forest and revenue departments were to then conduct a joint survey to arrive at a final decision.

The FRA of 2006 also provided for the recognition of individual and community forest rights. These rights included rights over occupied forest land. The first phase of RBS did not include forest dwellers. Subsequently, when the RBS was extended to forest *patta* holders also in 2018, beneficiaries had to be freshly identified. Disputes arose when a *patta* holder had died or sold the land (which was not permitted by law) or the *patta* was reported as lost.

6 | LRUP: THE AIMS AND THE OUTCOMES

According to the Telangana government, LRUP was a major success; an official report claimed that 'the programme has now been completed, thereby guaranteeing that land records data across the state is authentic' (GoT, 2022, p. 49). Elsewhere, the government also claimed that the 'purity' level of PPBs—i.e., the share of undisputed land records—increased from about 65% before the LRUP to 93% after the LRUP (Table 4). Only about 1 million acres of land were classified into Part B. A digitised LRMS—named *Dharani*—was also installed.[13] These achievements appeared commendable and tended to convey that LRUP successfully transcended the historically evolved constraints that the character of land posed to its accurate measurement.

Yet, a closer analysis flags the need for several serious qualifications and shows that the claims of the government were disconnected from the enormous unrest that had spread regarding the LRUP. To begin with, there were

[13]In 2011, the Revenue Department had begun using an online portal called *Webland*. However, as officials explained to us, *Webland* had no checks and balances. It allowed for entries to be randomly added or deleted. The tehsildars could add as many entries as they wished. This led to problems, such as a plot of land showing an area of 100 acres in the *sethwar* but 190 acres in *Webland*. *Webland* was replaced by an Integrated Land Records Management System (ILRMS) in 2018 and christened as *Dharani* in 2020. *Dharani* was supposed to have inbuilt checks and balances. Duplication of entries was eliminated by capping the area as indicated in the *sethwar*. It also integrated the silos of registration and revenue, so that the *tehsildar* would receive a notification immediately after a sale deed was registered.

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TABLE 4 Official figures on the final output of LRUP, Telangana, as of March 2019, in acres and percentage.

Item	Coverage (million numbers/acres)	Share (%)
Total extent verified, in million acres	*23.8*	*100.0*
Updated agricultural land, in million acres	15.8	66.4
Updated non-agricultural land and government assets, in million acres	7.0	29.4
Disputed land, in million acres	1.0	4.21
Total number of cleared agricultural khatas, in million numbers	*6.0*	*100.0*
Number of PPBs issued, in million numbers	5.6	93.1
Number of *khatas* cleared for PPBs, but not yet digitally signed, in million numbers	0.4	6.9

Abbreviations: LRUP, Land Records Updation Programme; PPBs, *Patta* Passbooks.
Source: Chief Commissioner of Land Administration Office, Revenue Department, Hyderabad.

widespread allegations of corruption in the LRUP. In July 2019, a tehsildar and a VRO in Ranga Reddy district were arrested for demanding a bribe from a farmer to correct his land records.[14] Videos of farmers falling on the feet of revenue officers and pleading them to correct erroneous land records became viral on social media.

Many attempts at suicides were also reported. In November 2019, a farmer allegedly attempted suicide by setting himself ablaze in Lingala Ghanpur Mandal in Jangaon; he alleged that his 1-acre plot was not registered in his name though he was cultivating it for many years.[15] In June 2020, a 63-year-old farmer allegedly committed suicide by consuming a pesticide in Peddapalli district; his suicide note contained a complaint against the tehsildar and the VRO.[16] In November 2019, a female MRO in Abdullahpurmet was allegedly killed by a landowner over a dispute over the ownership of a few acres of land.[17] In Karimnagar, petrol was thrown at revenue officers. Revenue officers also regularly received death and suicide threats. Gradually, complaints around the LRUP multiplied and threatened to metamorphose into a major political storm.

In 2021, the Telangana High Court chastised the state government for not responding to people's complaints. Following political agitations from opposition parties and interventions from the courts, the government formed a sub-committee of its Council of Ministers to study the problem.[18] In 2022, the sub-committee recommended that the LRUP must be succeeded by a 'digital field survey' to achieve a closure on the complaints—a decision that questioned the basis of the government's claim that land records had become 'authentic'. The *Dharani* portal was also 'a dead end', as a digital activist put to us; the sub-committee also recommended a substantial overhaul of the portal.

We proceed from here in the following manner. We first analyse the quantum and nature of complaints that arose out of LRUP and *Dharani* and the types of institutional interventions that were necessitated. We then attempt to understand the foundational features of LRUP that had a fundamental impact on its processes and outcomes. For this purpose, we make use of official data, data mobilised from helplines, personal interviews and media reports.

[14]See '93 Lakhs Cash, Jewellery Found at Home of Telangana Officer Who Won Award', *NDTV.com*, 12 July 2019, available at https://bit.ly/39RJNqk, accessed on 31 August 2022.
[15]See 'Vexed with Revenue Officials, Telangana Farmer Tries to Set Himself Ablaze', *The New Indian Express*, 12 November 2019, available at https://bit.ly/2ParWRV, accessed on 31 August 2022.
[16]See 'Officials "Tamper" Land Records, Farmer Ends Life in Telangana', *Times of India*, 21 June 2020, available at https://bit.ly/317tgKF, accessed on 31 August 2022.
[17]See 'Why Telangana Farmers Are Facing the Brunt of the New Digitised Land Record System', *The News Minute*, 20 December 2019, available at https://bit.ly/3giGTgF, accessed on 31 August 2022.
[18]'Three Cabinet Sub-committees Constituted', *The Hindu*, 16 September 2021, available at https://bit.ly/3R6uFtP, accessed on 31 August 2022.

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6.1 | The errors of measurement

Beginning from 2018, despite the guidelines set out to address specific disputes, there were a deluge of complaints from affected persons regarding measurement, recording and classification of land under the LRUP.

6.1.1 | Types of complaints

The first set of complains were related to Part B. Landowners complained that undisputed lands were classified in Part B; that despite numerous applications, the revenue authorities were hesitant to reclassify their plots; that applications for correction were rejected without citing any reason and the online status was just provided as 'Grievance Rejected'; that applications remained unresolved for months with the online status staying as 'Issue has been forwarded to the MRO Office'; and that for these reasons, they were not supplied with new PPBs. When aggrieved landowners went to courts, the government provided a standard response in its affidavits: that the government had notified the Telangana Rights in Land and Pattadar Passbooks Act 2020 and launched the *Dharani* portal; that it had prescribed various procedures under Section 6 in the Act to apply for mutation of the names; and that all such applications must be made online through the portal. Citing these responses from the government, lower courts dismissed petitions that were filed. If lower courts rejected most petitions, the Telangana High Court admitted some appeals and sent notices to the government.

The second set of complaints were on entries in Part A itself, which the government claimed as 'pure'. One controversy that attracted state-level attention was reported from four villages in Ranga Reddy district.[19] In 2015, the Telangana government had announced a 19,333-acre industrial project called 'Hyderabad Pharma City', which was to be managed by the Telangana State Industrial and Infrastructure Corporation (TSIIC). Although some farmers had given their consent for land acquisition for this project, large number of others had not. Yet, overnight, many non-consenting farmers found their names erased from the revenue records in *Dharani*; the new owner was listed as TSIIC. Officials dismissed the case as 'a technical error that occurred in the backend process'. An official entry in the *Dharani* portal admitted: 'even though a small extent of land is acquired by the government in a survey number, entire survey number is categorised as acquired due to mistake in entries.'[20]

However, following state-level furore that interpreted the 'technical error' as a conscious effort to intimidate landowners into consent, the court sent notices to the government. The sub-committee of the ministers was formed in the context of multiple controversies like in Ranga Reddy district. The sub-committee listed 20 different types of issues with LRUP and *Dharani*, which included (a) incorrect entries of the name and other details of the landowners; (b) wrong entry of private land as government land; (c) the problem of missing survey numbers in entries; (d) the problem of missing sub-divisions and some sub-divisions of a plot included under Part B; (e) cases of private land registered as temple land or prohibited land; (f) cases where one land parcel was registered under many names; and (g) software restrictions in *Dharani* on more than one person buying or selling land.[21]

There were more than 34,000 complaints related to missing survey numbers or missing sub-divisions, more than 16,000 complaints on incorrect mention of survey numbers in the PPBs and more than 100,000 complaints of sub-divisions classified as prohibited lands. Other reports, including in July 2022, noted that a total of more than 300,000 complaints were pending before the Revenue Department.[22] Some typical complaints were as follows[23]:

[19] Johnson, F. (2022), 'As Telangana Industrialises, Names of Hundreds of Farmers Disappear from a Govt Land-Records Database', *article 14*, 9 June, at https://tinyurl.com/57373mea, accessed on 31 August 2022.

[20] See entry in the *Dharani* portal at https://bit.ly/3pVWlWm, accessed on 31 August 2022.

[21] See 'Provision to Correct Errors on Dharani Soon', *Telangana Today*, 18 January 2022, available at https://bit.ly/3AXuNGu, accessed on 31 August 2022.

[22] See '"Dharani" May See Major Revamp', *The Hans India*, 8 July 2022, available at https://bit.ly/3wF58zB, accessed on 31 August 2022.

[23] See 'Dharani in the Court!', *Andhra Jyoti*, 5 September 2022, available at https://bit.ly/3xjYsaJ, accessed on 13 September 2022.

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In 1997, two brothers from Kankal village of Puduru mandal of Vikarabad district bought 2.07 acres of agricultural land from two people from the same village. They also received their PPB through mutation. Both the brothers who purchased the land died. 1.03 acres and 1.04 acres were registered in Dharani in the names of the wives of the brothers by fouthi. However, the land titles to this land of 2.07 acres were issued in the name of the daughter-in-law of one of the previous sellers ….

A man from Pulimamidi village, Kandukuru mandal, Ranga Reddy district, purchased two acres of land. He received his PPB before Dharani was instituted. When Dharani was instituted, it was the name of the previous seller that appeared as the landowner. Based on the Dharani record, that previous seller sold the land a second time to someone else. Currently, though the land is under possession of the purchaser in Pulimamidi, there are no title deeds for the same ….

A person from Gachibowli, Hyderabad bought 3.20 acres near Shankarpally, Ranga Reddy district. After Dharani's institution, he was issued a PPB for only two acres of land. A PPB for the remaining 1.20 acres was issued to another person from Shankarpally village …. When the actual buyer approached the revenue officers, he was told: 'we cannot do anything …. There is no module in Dharani to solve this problem …. Go to court'. When the complainant approached the person who received the PPB for the 1.20 acres, he demanded Rs 50 lakh to return the land ….

A key issue was also that all the earlier mechanisms of grievance redressal were disbanded with the inauguration of *Dharani*. In 2020, the post of *patwari* or the VRO was abolished. Thus, no land-related complaint could be raised or taken up at the village level. All complaints had to be made online in *Dharani* itself or another portal called *Mee Seva* (i.e., 'at your service'). Earlier, aggrieved persons could meet an official personally and submit, or appraise them of, their complaint. However, the new system eliminated all personal submissions or interactions. With no accountability on who was responsible for addressing the complaints, the complainants were left helpless even in cases where their names had simply disappeared from the records.

6.1.2 | Data from the *Kisan Mitra* helpline

Kisan Mitra (KM) was a telephone helpline for farmers run by the RSV. After the LRUP was conducted in Adilabad district, KM started receiving several complaints pertaining to land records. We analysed 475 land-related complaints lodged at KM between 8 February 2018 (when the LRUP was ongoing) and 8 May 2019. Out of the 475 complaints, 386 complaints were explicitly related to LRUP. These were grouped by us into a set of complaint types (see Table 5). We do not claim that the table provides an exhaustive list of the types or number of complaints on LRUP in Adilabad. However, it does provide insights into the diversity of complaints from farmers and their relative importance.

About 37% of the complaints were on not receiving the PPB. In most of these cases, callers were ignorant of the reason. About 20% of the complaints were on errors in area recorded under a survey number, or under survey numbers not appearing in the final 1-B register. About 12% of the complaints were on issues of succession over land. About 7% of the complaints were on incorrect details in the PPBs. Together, these four types constituted three-fourths of all the complaints.

We also analysed data from the KM helpline between November 2020 and August 2022, that is, after the *Dharani* portal was introduced. Five hundred ten complaints were received during this period, and the types of complaints revealed the range of issues that persisted: (a) errors of inclusion of undisputed lands under Part B; (b) errors in citing survey numbers and the problem of missing survey numbers; (c) errors in the extent of area cited in the

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TABLE 5 Distribution of types of complaints on LRUP received at the *Kisan Mitra* helpline, Adilabad district, February 2018 to May 2019, in number and percentage.

Type of land issue	Number of complaints	Share in total (%)
Did not receive the new digital PPB	141	36.5
Wrong area under a survey number/missing survey number	76	19.7
Complaints on succession on land	45	11.7
Incorrect details in the PPB	26	6.7
Area in dispute with forest department/ROFR *patta*	19	4.9
Disputes in assigned land (POT cases)	17	4.4
Title of the PPB changed without the owner's knowledge	10	2.6
Complaints related to land encroachment	10	2.6
Complaints related to mutation of land	13	3.4
Disputes in agency area (LTR issues)	6	1.6
Disputes in government acquired land	6	1.6
Complaints on *sadabainama* cases	6	1.6
Complaints related to landowner having committed suicide	2	0.5
Incorrect linking of PPB with *Aadhaar*	5	1.3
Calls demanding new land survey	3	0.8
PPB received, but landowner's name not registered online	1	0.3
Total	**386**	**100.0**

Abbreviations: LRUP, Land Records Updation Programme; LTR, Land Transfer Regulation; POT, Pot Kharab; PPB, *Patta* Passbook; ROFR, Record Of Forest Rights.
Source: Compiled from the helpline data.

PPB; (d) errors related to land mutation and land succession; (e) errors in listing the type of land; (f) errors in the name, caste and gender of the *pattadars*; (g) errors in classifying revenue land as forest land; (h) the omission of pathways to fields from land records; (i) errors in citing the extent of land acquired by the government for specific purposes; (j) absence of digital signature in the PPB; and (k) inclusion of revenue land as tribal land, and of tribal land as revenue land.

During our field work, we also encountered numerous other issues related to the *Dharani* portal. Some of the common complaints were (a) the system and data entry were slow; (b) the portal had too many runtime errors; (c) the photo of the PPB holder did not always appear in the portal; (d) despite different *khata* numbers, the first occupant's name appeared repeatedly in all the rows; (e) some corrections had to be approved by senior revenue officials, which were inordinately delayed; and (f) any decision of the *tehsildar* on a case where the area under a survey number exceeded the area in the *sethwar* had to be approved by the Joint Commissioner, which was often delayed.

The essential takeaway is that the claim of 'purity' of land records was illusory. A purity level of 93% only meant that new data were entered into 93% of the land records based either on the previous entries or on the responses received from persons whom the government assumed as the real landowners. The extensive nature of errors that remained in the system were evidence that any 'record purification exercise', without resorting to a resurvey of land, would only exacerbate complexities on the ground; as a leading advocate on land issues told us: 'corrections claimed to be made in the exercise were more or less nullified by the inclusion of new errors'.

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6.1.3 | The continuing stalemate

The sub-committee had suggested numerous measures to correct the errors in the database and in the portal design. As the post of *patwari* was abolished, district collectors were given the charge of addressing online complaints and authorising corrections. In cases of missing sub-divisions or survey numbers or the wrong registration of private land as other types of land, a new set of 'modules' were to be added to *Dharani*, and to spread awareness among people on the portal, a new help desk was to be initiated.

Though the government authorised the district collectors to resolve complaints, most district collectors were hesitant to take responsibility for correcting the errors. One collector told us that more than 80% of the complaints faced in their weekly public sessions—called *Prajavani*—were related to LRUP and *Dharani*, and in most of the cases, they had no jurisdiction or power to act. Even when they had jurisdiction, they refused to act in fear of future legal complications. 'We have been reduced to "*Dharani* Collectors"', the collector sarcastically remarked to us during an interview. According to another collector we interviewed, the new system gave no legal immunity to the collectors for the decisions they took; so, they just forwarded the file to the CCLA's office in the state capital. Reports also indicated that collectors had collectively asked the state government to relieve them of the responsibility of dealing with LRUP/*Dharani* issues and substitute it with an exclusive state-level cell of senior officials.[24]

But most importantly, the sub-committee admitted that the problems of missing survey numbers could not be resolved without a new digital field survey, that is, a fresh physical survey of land and entering the new survey numbers in all the PPBs.[25] Such a stance did not just raise the issue of whether the initial design of LRUP was robust. It effectively quashed the government's own earlier claims that the database had become 'authentic'. It also raised serious doubts on the bureaucratic confidence in the technocratic solution of *Dharani*.

7 | WHY DID THE ERRORS PERSIST?

The errors that accumulated in LRUP and *Dharani* were not purely administrative or technical in origin. We argue that these issues largely, also, originated from the design of the scheme that spilled over into the processes and outcomes. In this section, we discuss the important points of disjuncture between the aims of LRUP and the outcomes.

a. LRUP: Conclusive or presumptive titles?

What did the LRUP aim for? Did it aim to create a *conclusive* system or a *presumptive* system of land titling? It appears that Telangana chose neither; it chose a curious third way. On the one hand, conclusive titling needs a fresh resurvey of land. But Telangana's LRUP was not designed to be a resurvey; it was just an updating of textual records using a labour-intensive house-to-house ascertaining of whether the records held by households were 'close to the truth'. Only 3.6% of area in Telangana was resurveyed by April 2020.

On the other hand, elements of presumptive titling existed in the design of LRUP itself. For instance, sale deeds were extensively used to update textual land records with scant regard to the reality on the ground. The new land titles given were, largely, a derivative of the existing presumptive titles. The accumulated errors in the presumptive titles were transferred to the new 'purified' land records.[26]

In most Indian states, including Telangana, the administrative capability of governments to undertake a state-level

[24]See 'TS Collectors Seek Relief from Dharani Complaints', *Deccan Chronicle*, 20 January 2022, available at https://bit.ly/3TuHxeY, accessed on 31 August 2022.

[25]Telangana's design of LRUP may be contrasted with ongoing land records modernisation efforts in other states of India, such as Karnataka, Andhra Pradesh and Kerala. In 2022, these states had begun to make use of advanced technologies, such as CORS, in the resurveying of land to reduce the time and effort and increase accuracy. Although the success of these states remains to be examined in future research, they do represent alternatives to Telangana's chosen method.

[26]Bringing in a larger legal question into play, Goswami et al. (2020) have argued that 'without clear and accurate processes, it is unlikely that the courts will accept the change from presumptive to conclusive' titles.

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resurvey of land has been demonstrably poor (Saxena, 2005). In many states, government officials were hesitant to undertake fresh cadastral surveys (Narayanan et al., 2017). In Telangana too, there was scepticism among bureaucrats on the feasibility of a fresh resurvey. We asked a senior revenue official in Telangana in 2019 on why a fresh resurvey was not attempted under the LRUP. The answer was instructive:

> *A government cannot intervene in the personal land matters of farmers …. A cadastral survey in the whole State is time-consuming. It would also lead to the opening of a can of worms, which is totally uncalled for if the farmers and their neighbours are satisfied with the existing situation …. Suppose we did a resurvey, and the area with a farmer fell short by one acre compared to the revenue records. Will the government provide him that extra acre, for which he may have already paid for? If both X and Y are happy with the area written in their PPBs, regardless of their actual physical possession, no physical survey is required. A physical survey is required only when there is a dispute. For that, we have the Part B.*[27]

But clearly, the government's decision in 2022 to rely on a fresh digital field survey was evidence of the poor thinking behind the LRUP's design, which in turn led to poor outcomes.

b. LRUP: Land records modernisation or a welfare scheme?

LRUP was hurriedly designed in 2017 to ensure a time-bound introduction of the RBS. The then Telangana government led by the Telangana Rashtra Samiti (TRS) was facing a tough re-election in 2018. The deposit of a few thousand rupees into the bank accounts of farmers was seen by the TRS leadership as an ideal strategy to win rural support, particularly given that it was unsuccessful in addressing the long-term agrarian crisis in the state (GoAP, 2005; Ramachandran et al., 2010; Sainath, 2015).

Implementation of RBS required LRUP as a pre-requisite. As a result, the design of the LRUP was fundamentally different in its motivation, and its outcomes, from other efforts at land records modernisation. In an interview with us, S.K.—a lawyer with expertise on land rights—put it thus:

> *…. In the 1930s, the survey settlement and bandobust operations were done with the objective of who should pay the money. The LRUP was done with the objective of who should receive the money. This difference in the objective of the survey is indicative of the attitude with which it was carried out. (emphasis added)*

The specificity of the objective and tight timeline shaped not just the LRUP's design but also its effectiveness on the ground. In the interview, S.K. spoke to us about earlier pilot efforts to modernise land records.

> *We attempted such an exercise in eight villages in Warangal and Nalgonda districts with the participation of the village community. It took us six months to complete the entire process in eight villages, right from identifying a wrong entry to matching records to completing the legal processes to correct them. But in LRUP, they completed one village in just nine or ten days. You cannot even identify and list every problem in nine or ten days.*

Results from our field work corroborated this observation. The targets set were politically determined and overly ambitious. The time of 3 months (though extended by 3 more months) was inadequate to complete a massive

[27]The fear of whether a resurvey would be self-defeating is widespread within India's bureaucracy. As a result, alternative methods have often been experimented with by states, such as photogrammetry, use of handheld GPS devices and apps, electronic total stations (ETS) and satellite imagery, combined with ground-truthing (Sinha, 2017).

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exercise like the LRUP in more than 10,000 villages. Revenue functionaries we interviewed explained the nature of stress involved and how some staff fainted in the field. '*Aaj shaam tak ho jana chahiye*' ('the work should be completed by today evening') was a common instruction from the top. According to a *tehsildar*, *patwaris* collected data during the day, performed data entry during the night, slept in the office and woke up next morning to go straight to a *gram sabha*. Another revenue officer exclaimed that a step that required 7 days was ordered to be completed in 7 hours; we were told in an interview: 'the reality is that even if we got 10 months per village, instead of 10 days, ground-level verification could not have been satisfactorily completed.'

Lack of adequate manpower too affected the capacity of the government. The Revenue Department was considerably understaffed. Every village is supposed to have a *patwari*. In the Talamadagu *mandal*, there were 28 revenue villages but only 7 *patwaris*. *Tehsildars* complained about a shortage of trained surveyors too. Adilabad district had 18 *mandals*, for which there were only eight or nine surveyors. According to the district collector of Adilabad, 18 *mandals* should have had at least 18 surveyors because each *mandal* was further divided into 20–30 villages. Revenue officials also complained about the lack of modern tools to undertake cadastral surveys. As a result, resolutions of Part B cases across the state were considerably delayed.

c. The '*sethwar* problem'

The *sethwar* was the benchmark document in LRUP. Soon after the field exercises, freshly mobilised data had to be entered into the *Dharani* portal. The problem was the technocratic worldview built into the design of this portal. *Dharani* was so designed such that the total area under the new survey numbers had to compulsorily be equal to the corresponding areas listed in the *sethwar*, *khasra pahani* and the *pahani*. As one official remarked, 'even if the area was more or less by just one *gunta* than in the *sethwar*, the portal did not accept the data'.

A lack of correspondence of *sethwar* data and the ground truth should have been expected. After 1932–1934, the area, usage and ownership patterns in land had undergone major changes. There were large numbers of transactions, successions, mutations, partitions, exchanges, consolidation of parcels and encroachments. There were also numerous manipulations of land records by landlords and revenue officials.

We do not argue that the discrepancies between the *sethwar* and the ground truth in 2017 were irreconcilable. However, we do argue that the reconciliation for a whole village would have been difficult to be completed in just 10 days. In many cases, we were told that either the *tehsildar* exercised discretionary powers to decide on a dispute or such records were simply classified under Part B. It is from these discretionary powers that the large number of complaints regarding entries under both Part A and Part B may have arisen. Activists of the RSV gave us the example of a case in Venkatpetrao village in the Mancherial district; there, about 500 cases were classified as Part B in a hurry, just so that the tasks in the village could be completed on time. In fact, some reports indicated that about 200,000 cases of disputes were classified into Part B during the LRUP.[28]

Senior revenue officials who we interviewed were of the view that both LRUP and *Dharani* seriously underestimated the complexities in what was entered into the possessor column in land records. They were of the view that senior state-level officials did not listen to their suggestions that 'every entry in the possessor column required a ground-level verification'. However, their suggestions were disregarded. As *Dharani* was constructed as a 'top-down' system, with no possibilities of checks and balances or appropriate grievance redressals, every minor error or dispute also became vulnerable to scrutiny by the courts.

d. The exclusion of tenants

The exclusion of tenants from the land records was an important perverse outcome in the LRUP. According to the RSV's assessment, one third of all the annual payments under RBS went to absentee landlords.[29]

[28]See 'Lack of Central Funds Forces Govt to Give Up Land Survey', *Times of India*, 5 February 2019, Hyderabad, available at https://bit.ly/3TyY2Xh, accessed on 26 July 2020.

[29]See 'Winning Elections Telangana Style: Pay Farm Landowners, Ignore the Tiller', *Business Standard*, 20 December 2018, available at https://bit.ly/3SJDJpt, accessed on 30 September 2022.

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As per the Andhra Pradesh Record of Rights in Land and Pattadar Passbooks Act of 1971 (as amended in 1994), tenants, mortgagees and occupants of *Inam* land were eligible to apply for bank loans. However, Telangana amended this Act in 2020 and removed the need to record the interests of tenants in the PPB. As per Clause 3 of the 2020 Act, the RoR was supposed to contain only the name of the '*pattadars*'. So, the *Dharani* portal was programmed such that the owner column and the cultivator column in the new PPB had to have the same name as in the *pahani*. Only then would the digital PPB be generated. As a result, all the new PPBs distributed after LRUP mentioned just the name of the landowner.

The decision to exclude the tenants from RBS was a conscious policy measure of the Telangana government. In July 2018, the State's Chief Minister was quoted as saying that '*Rythu Bandhu* is meant for farmers. It is not tenant *Bandhu*'.[30] He further said: 'the government does not have details of tenant farmers. Nowhere in the official records have the details of tenant farmers been mentioned. Similarly, no farmer ever recognises a tenant farmer. The so-called tenant farmers will not have any right to the land.' During field work, an exasperated *tehsildar* in Adilabad district remarked to us: 'our Chief Minister does not want tenancy in Telangana!'[31]

We asked the district-level agricultural officers in Adilabad on why the beneficiaries of RBS were selected not based on who cultivated the land but based on whose name appeared on the land records. According to one officer, the RBS was 'meant to encourage farming so that people do not leave their farms and the phenomenon of absentee landlordism ends. It helps to induce big farmers to continue farming by providing them with monetary assistance. We want agriculture to grow and that is why it is important to give such assistance to the big farmers'. Clearly then, the RBS was loaded heavily against the interests of tenants in its design itself.

Criticisms on the exclusion of tenants have met with the response that absentee landlords would ultimately share RBS benefits with the tenants and lease rates will gradually fall (Mandala & Murty, 2021). According to a view, 'if the scheme is advertised properly, the market should adjust the tenant rents automatically'.[32] We conducted a sample survey of agricultural households in Talamadagu village in May 2019; in the sample, 35 households were tenants (see Ramesh, 2020). Not a single absentee landowner was found to share their RBS benefits with tenants. About 98% of the tenants said that their lease rates had either increased or remained the same after RBS. About 60% of the tenants said the lease rates had risen after RBS.

The exclusion of tenants from land records also led to the denial of multiple public services to farmers—most importantly bank credit. According to Clause 12 in the 2020 Act, 'any credit agency shall grant loan on the basis of Record of Rights maintained electronically'. As the RoR or *Dharani* had no provision to record the tenant's name, many tenants were denied access to bank credit. Such an exclusion was a violation of the Land Licensed Cultivators Act of 2011, which made bank credit an entitlement of the tenants. Agricultural extension services and several mobile applications to assist farmers were also stacked on to the *Dharani* database; because tenants did not have their names in *Dharani*, these services were also denied to them. Benefits from other cash transfer and crop insurance schemes like *Rythu Bima* and PM Kisan were also denied to tenants in many regions.

e. Subversion by local actors

A major complaint heard during our field work and in interviews with farmer's organisations was that corruption was rampant in the settlement of disputes. These complaints were largely levelled at officers of the Revenue Department at the local level. The abolition of the *patwari*, the insistence that all complaints be made online and the authorisation given to district collectors to resolve them were supposed to be measures to reduce corruption and manipulation of records.

However, the district collectors we interviewed were of the view that the new system retained, if not expanded, the possibilities of corruption. The collectors had no prior knowledge of the intricacies of each complaint that

[30]'Rythu Bandhu Scheme Not to be Extended to Tenant Farmers: CM K Chandrasekhar Rao', *The New Indian Express*, 1 July 2018, available at https://tinyurl.com/34pt3jjn, accessed on 30 September 2022.

[31]See also RSV and TISS (2018).

[32]See interview with Ashok Gulati, '*Rythu Bandhu* a Much-Superior Policy Than Even Loan Waivers', *The Print*, 12 December 2018, available at https://bit.ly/2LgxsQg, accessed on 30 September 2022.

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reached them from every corner of their district. These complaints could not be disposed of without asking for inputs from the very same *patwaris* or tehsildars whom the new system was supposed to have displaced. This left open the possibilities of manipulation or misrepresentation. In addition, the new powers given to the collectors opened a new avenue of corruption. A. Prasad Rao, a leader of the *All India Kisan Sabha*, told us in an interview that:

> *In the Kisan Sabha, we have recorded numerous reports of corruption at the ground level even for discharging legitimate duties. This was particularly evident during the second tranche of issuing PPB. Data entry was entrusted to contract workers (data entry operators), who had neither stake in nor commitment to their tasks. Huge amounts of money were demanded and accepted either directly by them or through middlemen. Many of these cases were reported in the media. But no prompt action was taken by the government. There were some suicides or attempts at suicide by the parties adversely affected by the delays in obtaining PPB. There were also some physical attacks on the concerned officers.*

8 | CONCLUDING COMMENTS

Land is an economic entity embedded in the social, political and cultural specificities of rural areas. The nature of land distribution influences social relations and shapes local power structures. Hence, the need for formal, accurate and updated records of land titles has been an important policy concern in the developing countries. Modern land records are necessary to further redistributionist agrarian goals, reduce exploitation of poor peasants, expand transparency guarantees and facilitate scientific land use planning. This paper was an assessment of efforts to modernise land records in India, with special reference to an effort to update textual land records in one state, Telangana.

Despite the strong case that exists for modernising land records, our overall conclusions highlight the tensions that surround the politics of land in rural societies. Given the strong landlord-led resistance to land reforms, land administration was always a vexed question in the post-War development project. However, the fundamentals of land administration shifted under neoliberalism; governments eschewed egalitarianism, discarded land reforms and embraced the goal of free land markets. Modernisation of land administration also became a tool to further technocratic goals of neoliberal policy, such as digitalised and targeted cash transfers.

Yet, as this case study in Telangana illustrates, the nature of political economy around land can frustrate attempts by the neoliberal state to dislocate land from the bundle of claims and counterclaims that have survived and accumulated over centuries and 'fix' conclusive land titles. Although the LRUP was a commendable effort, the Telangana government's claim of having 'purified' 93% of the land titles was far from the reality on the ground. There were numerous complaints on the accuracy of the new land records, which snowballed into political protests, violence and interventions by courts. By 2022, the government admitted to multiple issues with the project and its online land management portal—*Dharani*—and recommended a fresh digital survey.

The problem, we argue, is that the neoliberal state envisages land records modernisation as an erasure of history, in disregard of egalitarianism in land distribution and under the assumption that the 'illegibility' of records is a technical problem of administrative dysfunction. Aided by technocracy, the neoliberal state seeks to flatten out the historical multiplicities of claims and counterclaims on land; in their place, it superimposes a new system, actively intermediated by poorly designed portals, which essentially puts in place a new set of claims inadequately vetted by democratic exercises and laced with a new logic of power. The complete exclusion of tenants from the LRUP in Telangana is an illustration. All grievances hit a dead end, with an impersonalised portal absorbing complaints with no accountability to respond—either effectively or in time.

The outcomes of the LRUP reflect the continuing role of land as a marker of rural power. A lesson, surely, is that public policy on land administration must imbibe a better appreciation of broader historical and political

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considerations surrounding the ownership and use of land. Other Indian states plan to use alternative methods, including satellite and drone technologies, to fix problems in land records. It remains to be seen if they would succeed in circumventing the systemic constraints that the Telangana experiment encountered.

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CONFLICT OF INTEREST STATEMENT

We declare that we do not have any conflict of interest with regard to this paper.

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REFERENCES

Alm, J., Annez, P., & Modi, A. (2004). *Stamp duties in Indian states: A case for reform*", Policy Research Working Paper 3413. The World Bank. https://doi.org/10.1596/1813-9450-3413

Alston, L. J., Libecap, G. D., & Mueller, B. (1999). *Titles, conflict, and land use: The development of property rights and land reform on the Brazilian Amazon frontier*. University of Michigan Press. https://doi.org/10.3998/mpub.16208

Appu, P. S. (1996). *Land reforms in India: A survey of policy, legislation and implementation*. Vikas Publishing House.

Atwood, D. A. (1990). Land registration in Africa: The impact on agricultural production. *World Development*, *18*(5), 659–671. https://doi.org/10.1016/0305-750X(90)90016-Q

Bal, M. (2017). *Securing property rights in India through distributed ledger technology*. Observer Research Foundation.

Bell, K. (2006). World Bank support for land administration and management: Responding to the challenges of the Millennium Development Goals. Paper presented at the XXIII FIG Congress, Munich, Germany, 8–13 October.

Benjamin, S., & Raman, B. (2011). Illegible claims, legal titles, and the worlding of Bangalore. *Revue Tiers Monde*, *2*(206), 37–54. https://doi.org/10.3917/rtm.206.0037

Bobbili, A., & Manohar, T. (1999). Failure of Record of Rights in land in Telangana: A case study of a Waddepally Village. *Proceedings of the Indian History Congress*, *60*, 877–884.

Bobbili, A., & Manohar, T. (2001). Role of the 'Patwari' in preparation of revenue records in Telangana. *Proceedings of the Indian History Congress*, *62*, 562–567.

Burman, A. (2019). Making land titles in India marketable: Using title insurance as a viable alternative to conclusive titling. *Washington International Law Journal*, *28*(1), 109–135.

Byamugisha, F. F. K. (1999). The effects of land registration on financial development and economic growth: A theoretical and conceptual framework. World Bank Policy Research Working Paper 2240, Washington, November. https://doi.org/10.1596/1813-9450-2240

Byres, T. J. (1988). The agrarian question and the differing forms of capitalist agrarian transition: An essay with reference to Asia. In J. Breman & S. Mundle (Eds.), *Rural transformation in Asia*. Oxford University Press.

CAG. (2016). *Report of the comptroller and auditor general of India on revenue sector*. Government of Telangana.

Capps, G. (2016). Tribal-landed property: The value of the chieftaincy in contemporary Africa. *Journal of Agrarian Change*, *16*(3), 452–477. https://doi.org/10.1111/joac.12179

Cole, M. (2022). *What's wrong with technocracy*. Boston Review. August.

Daksh. (2016). *Access to justice survey 2015–16*. Daksh India.

De, G. (2016). Whose land is it anyway? Property, law and rights in the land acquisition question in India. *Journal of Land and Rural Studies*, *4*(1), 23–35. https://doi.org/10.1177/2321024915616670

Deininger, K., & Feder, G. (2013). Land registration, governance, and development: Evidence and implications for policy. *The World Bank Research Observer*, *24*(2), 233–236.

Dong, X. Y. (1996). Two-tier land tenure system and sustained economic growth in post-1978 rural China. *World Development*, *24*(5), 915–928. https://doi.org/10.1016/0305-750X(96)00010-1

14710366, 2023, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/joac.12556 by Tata Institute Of, Wiley Online Library on [06/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Feder, G., & Nishio, A. (1998). The benefits of land registration and titling: Economic and social perspectives. *Land Use Policy*, *15*(1), 25–43. https://doi.org/10.1016/S0264-8377(97)00039-2

GoAP. (2005). *Report of the commission on farmer's welfare*. Government of Andhra Pradesh.

GoI. (1961). *Report of sub-committee on land management practices*. Land Reforms Implementation Committee, Government of India.

GoI. (2009). *Report of the committee on state agrarian relations and the unfinished task in land reforms*. Department of Land Resources, Ministry of Rural Development, Government of India.

Goswami, A., Jha, D., & Lushington, K. (2020). Approaches and methods of land title legislation in India: Far from reality or close to the ground. In V. Bhagat-Ganguly (Ed.), *The land question in neoliberal India: Socio-legal and judicial interpretations*. Routledge. https://doi.org/10.4324/9781003042389-3

GoT. (2022). *Telangana socio economic outlook 2022*. Planning Department, Government of Telangana.

Gupta, P. S. (2011). Ending finders, keepers: The use of title insurance to alleviate uncertainty in land holdings in India. *UC Davis Journal of International Law and Policy*, *17*, 63–109.

Karmakar, K. G., Banerjee, G. D., & Mohapatra, N. P. (2011). *Towards financial inclusion in India*. Sage Publishers.

Khusro, A. M. (1958). *Economic and social effects of Jagirdari abolition and land reforms in Hyderabad*. Department of Publications and University Press, Osmania University.

Lenin, V. I. (1964). New data on the laws governing the development of capitalism in agriculture. In *Lenin's collected works* (Vol. 22) (pp. 13–102). Progress Publishers.

Levien, M. (2018). *Dispossession without development: Land grabs in neoliberal India*. Oxford University Press.

Mandala, S. R., & Murty, C. S. (2021). Land lease market under irrigated and rainfed conditions in Andhra Pradesh and Telangana. *Indian Journal of Agricultural Economics*, *76*(2), 261–277.

Marx, K. (1853). *India*. New York Daily Tribune. August 5.

Marx, K. (1959). *Capital* (Vol. 3). International Publishers.

McKinsey. (2001). *India: The growth imperative*. McKinsey Global Institute.

Miceli, T., & Kieyah, J. (2003). The economics of land title reform. *Journal of Comparative Economics*, *31*(2), 246–256. https://doi.org/10.1016/S0147-5967(03)00041-6

Mittal, A. (2014). *Is the World Bank enabling agribusiness land grabs?* Al Jazeera. 12 April.

Nagaraj, R. (2019). *Make in India: Why didn't the lion roar?* The India Forum. 16 May.

Narayanan, S., Prabhakar, P., Shaikh, G., Uday, D., & Zaveri, B. (2018). *Digital India land records modernisation programme: Assessing impact in Himachal Pradesh and Maharashtra*. Ideas for India. 6 August.

Narayanan, S., Shaikh, G., Uday, D., & Zaveri, B. (2017). *Report on the implementation of the DILRMP in the state of Maharashtra*. Finance Research Group, ndira Gandhi Institute of Development Research.

Nayak, P. (2020). Land titling or land reforms: India's policy dilemma. In D. K. Mishra & P. Nayak (Eds.), *Land and livelihoods in neo-liberal India* (pp. 35–56). Palgrave Macmillan. https://doi.org/10.1007/978-981-15-3511-6_2

NCA. (1976). *Report of the National Commission on Agriculture*. Ministry of Agriculture and Irrigation, Government of India.

NCAER. (2017). *A pilot impact assessment of the digital India land records modernisation programme*. National Council of Applied Economic Research, Indira Gandhi Institute for Development Research and National Institute of Public Finance and Policy.

NCAER. (2020). *The NCAER land records and services index*. National Council of Applied Economic Research.

Panicker, V. (2015). *Land records modernisation: Technology transitions*. Indian Institute for Human Settlements.

Planning Commission. (1973). *Report of the task force on agrarian relations*. Government of India.

Polanyi, K. (1944). *The great transformation*. Beacon Press.

Ramachandran, V. K. (2020). *Modernization and digitization of land records in Kerala: A report and recommendations*", Unpublished Note. Kerala State Planning Board.

Ramachandran, V. K., Rawal, V., & Swaminathan, M. (2010). *Socioeconomic surveys of three villages in Andhra Pradesh*. Tulika Books.

Ramesh, P. (2020). *Why cash transfers cannot replace structural reforms in agriculture*. Indiaspend. 23 June.

Rao, P. T. (2014). *Status of Adivasis: Andhra Pradesh, Indigenous peoples land series*. Aakar Books.

RSV and TISS. (2018). *Farmer suicides, land ownership, tenant farmers and Rythu Bandhu*. Rythu Swarajya Vedika and Tata Institute of Social Sciences.

Sainath, P. (2015). *The slaughter of suicide data*. Frontline. 21 August.

Saxena, N. C. (2005). Updating land records: Is computerisation sufficient? *Economic and Political Weekly*, *40*(4), 313–321.

Scott, J. C. (1998). *Seeing like a state: How certain schemes to improve the human condition have failed*. Yale University Press.

Sen, A. (1999). *Development as freedom*. Oxford University Press.

Sinha, R. (2017). Adapting the Torrens conclusive titling system in India: Debatable issues and way forward. In V. Ganguly & S. Mishra (Eds.), *Journey towards land titling in India* (pp. 66–85). The Lal Bahadur Shastri National Academy of Administration.

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de Soto, H. (2000). *The mystery of capital: Why capitalism triumphs in the west and fails everywhere else*. Basic Books.
Srivastava, B. B. (2015). Land governance: Issues, challenges and way forward. In V. Bhagat-Ganguly (Ed.), *Land rights in India: Policies, movements and challenges*. Routledge.
Sud, N. (2021). *The making of land and the making of India*. Oxford University Press.
Sundarayya, P. (1972). *Telangana people's struggle and its lessons*. Communist Party of India (Marxist).
Uday, D., & Zaveri, B. (2020). *Linking welfare distribution to land records in India: Part 1*. The Leap Blog. 14 March.
Wadhwa, D. S. (1989). *Guaranteeing title to land: A preliminary study*. N. M. Tripathi Pvt. Ltd.
Wadhwa, D. S. (2002). Guaranteeing title to land. *Economic and Political Weekly*, *37*(47), 4699–4722.
World Bank. (2007). *India: Land policies for growth and poverty reduction*. World Bank.

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